

The table below summarizes the results of a new test to identify patients with an acute myocardial infarction (AMI).

The test was performed on 400 subjects.

Acute Myocardial Infarction			
Test Result	Present	Absent	Total
	Positive	120	30
	Negative	80	170
Total	200	200	400

If one of the 400 subjects in the study is randomly chosen, what is the likelihood that an abnormal test indicates that the individual has an AMI?

- A. 0.50
- B. 0.60
- C. 0.68
- D. 0.80
- E. 0.85

Explanation:

The correct answer is D. The likelihood of an abnormal test indicating disease, or the positive predictive value of a test (PV+), is expressed with the following formula:

$$PV+ = TP/(TP + FP)$$

where a true positive (TP) is an abnormal test result in an individual with disease, while a false positive (FP) is an abnormal test result in a healthy person. Note that as the FP rate decreases, the PV+ approaches 100%. The FP rate reflects the test's specificity, or the chance that a test is normal in a healthy individual. As the FP rate of a test decreases, the test's specificity approaches 100%. Therefore, a test with 100% specificity always has a PV+ of 100%. This underscores why tests with high specificity are used to confirm disease, since an abnormal test is

more likely to represent a TP rather than a FP.

From the table we can see the following: TP = 120, FN = 80, TN = 170, and FP = 30. The likelihood of an abnormal test indicating an AMI is therefore:

$$PV+ = 120/(120 + 30) = 0.80$$

Thus there is an 80% chance that an abnormal test is a TP and a 20% chance that it is a FP.

Choice A is the prevalence of AMI in this study. Prevalence refers to the number of subjects with disease (TP +

FN) divided by the total number of subjects in the study. Hence, the prevalence of an AMI in this study is $(120 +$

$80)/(120 + 80 + 170 + 30) = 0.50$, or 50%. As expected, the higher the prevalence of disease, the greater the

PV+, since the number of individuals with disease (TPs) is increased.

Choice B is the test's sensitivity $[TP/(TP + FN)]$, which is calculated as follows: $120/(120 + 80) = 0.60$, or 60%.

The test's sensitivity is too low to be useful as a screening test for an AMI.

Choice C is the negative predictive value $(PV-) = [TN/(TN + FN)]$, which is calculated as follows: $170/(170 + 80) =$

0.68, or 68%. This percentage indicates that there is a 68% chance that a normal test is a TN and a 32% chance

that it is a FN. The higher the prevalence of disease, the lower the PV-, since a greater number of FNs are present.

Choice E is the test's specificity $[TN/(TN + FP)]$, which is calculated as follows: $170/(170 + 30) = 0.85$, or 85%.

Although this test's specificity is higher than its sensitivity, its specificity is not high enough for it to be useful as a

confirmatory test for an AMI.

Trisomy 21

	Normal karyotype
Positive test	
100	
	50
Negative test	
100	
	250

An experimental diagnostic test is developed to noninvasively detect the presence of trisomy 21, Down's

syndrome. The test is administered to a group of 500 women considered to be at risk for a Down's fetus based on

blood tests. The results of this test are shown above. What is the sensitivity of this new test?

- A. 40%
- B. 50%
- C. 67%
- D. 71%
- E. 83%

Explanation:

The correct answer is B. Sensitivity is defined as the ability of a test to detect the presence of a disease in those who truly have the disease. It is calculated as the number of people with a disease who test positive (true positive) divided by the total number of people who have the disease (true positive + false negative). In this case, sensitivity equals the number of babies born with trisomy 21 whose mothers tested positive (100) divided by the total number of babies born with trisomy 21. This yields $100/200 = 50\%$ (not a very sensitive test).

40% (choice A) corresponds to the prevalence of the disease in the tested population, which in this case equals the total number of babies with Down's syndrome (true positive + false negative = $100 + 100 = 200$) divided by the total number of people tested (500). This yields $200/500 = 40\%$.

67% (choice C) corresponds to the positive predictive value of the test, which equals the number of babies with Down's whose mothers test positive (true positives = 100) divided by the total number of mothers testing positive (true positive + false positive = 150). This yields $100/150 = 67\%$.

71% (choice D) corresponds to the negative predictive value of the test, which equals the number of normal babies whose mothers tested negative (250) divided by the total number of people testing negative (350). This yields $250/350 = 71\%$.

83% (choice E) corresponds to the specificity of the test, which equals the number of babies without Down's whose mothers tested negative (250) divided by the total number of babies without Down's (300). This yields $250/300 = 83\%$.

An easy way to remember these concepts is:

Sensitivity = true positives/all diseased

Specificity = true negatives/all normal

PPV = true positives/all positives

NPV = true negatives/all negatives

Prevalence = all diseased/total population

A particular association determines membership on the basis of members' IQ scores. Only those persons who have documented IQ scores at least 2 standard deviations above the mean on the Wechsler Adult Intelligence Scale (WAIS) are eligible for admission. Of a group of 200 people randomly selected from the population at large, how many would be eligible for membership to this society?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Explanation:

The correct answer is E. 95% of a normally distributed population will fall between plus or minus 1.96 standard deviations from the mean. Since the population is normally distributed with regard to IQ, this means that

approximately 2.5% of the population will have IQ scores 2 standard deviations or more above the mean, and
2.5% of the population will have IQ scores 2 standard deviations or more below the mean. 2.5% of 200 people
is 5 people.

An urban health department reports the infant mortality rate of the city in the past year. This value corresponds
to which of the following?

- A. The number of deaths during the past year from all causes divided by the total population
- B. The number of deaths occurring after the first 4 weeks of life and before the first birthday divided by the total number of live births
- C. The number of deaths occurring until the first birthday divided by the total number of live births
- D. The number of deaths occurring within the first 4 weeks of life divided by the total number of births
- E. The number of deaths occurring within the first 4 weeks of life divided by the total number of live births

Explanation:

The correct answer is C. This is one of those questions where predicting the correct answer will facilitate your selecting the appropriate answer choice. Make sure to read each choice carefully (for example, choices D and E, though both incorrect in this case, look almost identical and could cost you points if you were not reading actively).

Choice A refers to the crude mortality rate.

Choice B refers to the postnatal mortality rate.

Choice D is an incorrect rendering of the neonatal mortality rate since it considers the total number of births instead of the total number of live births.

Choice E refers to the neonatal mortality rate.

Researchers are assessing the specificity of a screening procedure for breast cancer in a population of 3000

women whose mothers had the disease. The presence or absence of a malignant condition is established by

conventional mammography, which is assumed to be definitive for the purposes of comparison. The following data

are collected:What is the specificity of the test?

	Positive	Negative	Total
Cancer	90	10	100
No cancer	360	2540	2900
Total	450	2550	3000

A. $90/100$

B. $90/450$

C. $2540/2550$

D. $2540/2900$

E. $(90+2540)/3000$

Explanation:

The correct answer is D. Specificity refers to how well a test identifies persons who do not have the disease in question. In the example, 2900 women do not have breast cancer, and 2540 of these women have a negative test. Specificity is given by $(\text{true negatives}) / (\text{true negatives} + \text{false positives}) = 2540 / (2540 + 360) = 2540 / 2900$.

Choice A, $90/100$, represents the sensitivity of the test; that is, the proportion of women with the condition who test positive.

Choice B, $90/450$, gives the predictive value of a positive test. Of the 450 women who tested positive, 90 actually had the condition.

Choice C, $2540/2550$, gives the predictive value of a negative test. Of the 2550 who tested negative, 2540 did not have cancer.

Choice E, $(90 + 2540) / 3000$, refers to the efficiency of a test; that is, the proportion of all subjects who were correctly classified by the test.

An experimental serologic test is developed to detect the presence of HIV antibody. Epidemiological analysis reveals the results shown below.

	People with HIV antibody	People without HIV antibody
Positive test	100	50
Negative test	20	950

What is the sensitivity of this test?

- A. 11%
- B. 67%
- C. 83%

D. 95%

E. 98%

Explanation:

The correct answer is C. Sensitivity is defined as the ability of a test to detect the presence of a disease in those who truly have the disease. It is calculated as the number of people with a disease who test positive (true positive) divided by the total number of people who have the disease (true positive + false negative). In this case, sensitivity equals the number of people with HIV antibody who test positive (100) divided by the total number of people who have HIV antibody (120). This yields $100/120 = 83\%$ (not a very sensitive test).

11% (choice A) corresponds to the prevalence of the disease in the tested population, which in this case equals the total number of people with HIV antibody (true positive + false negative = $100 + 20 = 120$) divided by the total number of people tested ($100 + 20 + 50 + 950 = 1120$). This yields $120/1120 = 11\%$.

67% (choice B) corresponds to the positive predictive value of the test, which equals the number of people with the disease who test positive (true positives = 100) divided by the total number of people testing positive (all positives = 50). This yields $100/150 = 67\%$.

95% (choice D) corresponds to the specificity of the test, which equals the number of people without HIV antibody who test negative (950) divided by the total number of people without HIV antibody (1000). This yields $950/1000 = 95\%$.

98% (choice E) corresponds to the negative predictive value of the test, which equals the number of people without HIV antibody who test negative (950) divided by the total number of people testing negative (970). This yields $950/970 = 98\%$. An easy way to remember these concepts is:

Sensitivity = true positives/all diseased

Specificity = true negatives/all normal

PPV = true positives/all positives

NPV = true negatives/all negatives

Prevalence = all diseased/total population

If you prefer charts and formulas:

	(Diseased)	(Normal)
	People with HIV antibody	People without HIV antibody
Positive test	a, true pos.	b, false pos.
Negative test	c, false neg	d, true neg.

Sensitivity = $a/(a + c)$

Specificity = $d/(b + d)$

PPV = $a/(a + b)$

NPV = $d/(c + d)$

Prevalence = $(a + c)/(a + b + c + d)$

Choose your favorite approach. Just make sure you know this stuff, as it is almost always tested and gains you easy points once you have mastered the concepts.

A researcher publishes new results on the efficacy of a procedure using transplanted embryonic cells to correct a diabetic condition induced in experimental animals by chemical ablation of pancreatic beta cells. At another institution, a different researcher attempts to reproduce the findings, but cannot. This disparate finding calls into question the first study's

- A. accuracy
- B. power
- C. reliability
- D. sensitivity
- E. validity

Explanation:

The correct answer is C. The reliability of a measurement refers to the ability of other researchers to reproduce the results. This can be summarized by noting that reliability = reproducibility.

The accuracy (choice A) of a measurement refers to the "trueness" of the measurement, i.e., how close the measured value is to the true value. It is a measure of bias. Just because another researcher could not reproduce the findings does not necessarily imply that they were inaccurate.

The power (choice B) of a study is best conceptualized as the probability of rejecting a null hypothesis that is in fact false. This is equal to 1 minus the probability of a type II error ($1 - \beta$), since β is the probability of failing to reject a false null hypothesis.

The sensitivity (choice D) of a test is a measure of its ability to detect the presence of a disease in those who truly have the disease.

The validity (choice E) of a measurement is an index of how well the test measures what it purports to measure.

The disparate results obtained by another researcher do not necessarily mean that the results are not valid.

A group of researchers mistakenly conclude from a poorly designed experiment that acetaminophen cures the common cold. They have committed which of the following?

- A. $1 - \beta$ error
- B. Alpha error
- C. Beta error
- D. Type I error
- E. Type II error

Explanation:

The correct answer is D. Type I errors occur when researchers reject the null hypothesis when they should not have. In other words, they conclude a significant result when in actuality it does not exist. This is a particularly dangerous error to make, as it could lead to the administration of an ineffective drug to patients in need of life-preserving treatment.

The term " $1 - \beta$ error" (choice A) is not used, but $1 - \beta$ does correspond to a meaningful value. It reflects the power of a study, which equals the probability of rejecting the null hypothesis when it is false, or in other words, the probability of discovering a true relationship (e.g., concluding that acetaminophen has antipyretic action).

The term "alpha error" (choice B) is not used, but alpha values do correspond to a meaningful concept. Alpha is the probability of committing a type I error (explained above). Alpha corresponds to the p value ($< .05$) commonly used in statistical analysis.

The term "beta error" (choice C) is not used, but beta values do correspond to a meaningful concept. Beta is the probability of committing a type II error. Type II errors (choice E) occur when researchers accept the null hypothesis when it is false, thereby failing to detect a true relationship (e.g., concluding that acetaminophen has no antipyretic action).

A good mnemonic for distinguishing type I and type II errors is: Type I kills everyone. Type II makes professors blue (because they can't publish their statistically insignificant results).

A group of patients with lung cancer is matched to a group of patients without lung cancer. Their smoking habits over the course of their lives is compared. Based on this information, researchers compute the rate of lung cancer in patients who smoke versus those who never smoked. This is an example of a

- A. case-control study
- B. cohort study
- C. cross-sectional study
- D. longitudinal study
- E. randomized controlled study

Explanation:

The correct answer is A. Case-control studies are retrospective and are as described in the question stem.

Case-control studies allow researchers to compute an odds ratio.

In cohort studies (choice B), subjects are assembled on the basis of some common experience (such as attending medical school) and are then monitored for a specified amount of time at regular intervals (e.g.,

taking USMLE Steps 1, 2, and 3; see also longitudinal studies below) until they develop the outcome of interest (they become practicing physicians) or the follow-up time ends. The cohort study minimizes many of the biases

evident in case-control designs and is the definitive observational clinical study.

Cohort studies allow researchers to compute a relative risk.

Cross-sectional studies (choice C) usually have more modest goals than those of case-control and cohort

studies. A variable or group of variables is measured in a sample of a larger population to get an idea of the distribution and interrelationships of those variables in that population.

Longitudinal studies (choice D) identify individual subjects and follow them over a given period of time. For example, the study of cholesterol-lowering drugs on cardiovascular events requires that the same subject is observed over a significant period of time (e.g., 10 years).

A randomized controlled trial (choice E) is considered the most rigorous and powerful approach to answering a clinical question in which two treatments, strategies, or therapies are compared or when one therapy is compared to placebo. In this type of study, subjects are assigned treatments on a randomized basis.

A new drug with in vitro activity against HIV is tested on a population of patients with Western-blot confirmed HIV infections. Out of the 200 individuals in the patient population, 100 individuals are chosen by lottery to receive the drug. The drug, which is tasteless, is administered in a cup of orange juice; the other patients receive pure orange juice. Neither the nurses, doctors, nor patients know which patients receive the drug. At the end of the study period, the number of CD4+ T cells is determined for all of the subjects. This is an example of a

- A. case-control study
- B. case report
- C. cohort study
- D. cross-sectional study
- E. double-blind randomized clinical trial

Explanation:

The correct answer is E. This is an example of a placebo-controlled, double-blind randomized clinical trial. In this type of study, subjects with a particular disorder are randomized (the lottery) to receive either the treatment in question or a placebo. The information about whether the patient receives the treatment or the placebo is not known to either the subject or the investigator (double-blind). This type of study may be used to infer causality,

i.e., if the patients taking the medication have more CD4⁺ T cells than the other group, it is due to the drug in question.

A case-control study (choice A) is a retrospective study that pairs known cases of a disease with matched controls. This type of study gives information about the importance of risk factors. The lack of a prospective design makes inferences about causality difficult.

A case report (choice B) is a published report of a single incidence of a disease in a subject. This is usually done for rare diseases, or for rare associations of a disease with a particular risk factor.

A cohort study (choice C) is an observational (non-interventional) study that tracks subjects prospectively. This type of study gives a measure of incidence of a disease.

A cross-sectional study (choice D) samples at a selected point in time, matching individuals with the disease with controls without the disease. The lack of a prospective design makes inferences about causality difficult.

In a city with a population of 1,000,000, 10,000 individuals have HIV disease. There are 1000 new cases of HIV disease and 200 deaths each year from the disease. There are 2500 deaths per year from all causes. Assuming no net emigration from or immigration to the city, the incidence of HIV disease in this city is given by which of the following?

A. $200/1,000,000$

B. $800/1,000,000$

C. $1000/1,000,000$

D. $2500/1,000,000$

E. $10,000/1,000,000$

Explanation:

The correct answer is C. The incidence of a disease is given by the number of new cases in a given period divided by the total population. In this case, this is equal to $1000/1,000,000$.

The disease-specific mortality rate is the number of deaths per year from a specific disease divided by the population; in this case, $200/1,000,000$ (choice A).

The rate of increase of a disease is given by the number of new cases per year, minus the number of deaths (or cures) per year, divided by the total population. Since there is yet no cure for HIV disease, the number of cures is 0. In this case, the rate of increase is $(1000-200)/1,000,000 = 800/1,000,000$ (choice B).

The crude mortality rate is given by the number of deaths from all causes, divided by the population; in this case, $2500/1,000,000$ (choice D).

The prevalence of a disease is defined as the number of cases of the disease at a given time divided by the total population. In this case, it would be $10,000/1,000,000$ (choice E).

A pharmaceutical researcher is examining the ulcerogenic potential of a new nonsteroidal anti-inflammatory drug.

He gives 20 rats a single subcutaneous injection of the drug every day for one week, and gives a similarly matched group of 20 animals daily saline injections for one week. 24 hours after the final injection, the investigator sacrifices the rats, removes their stomachs, and examines them to determine if any ulcers were produced. He obtains the following data:

		Ulcers present	Ulcers absent
Drug	12		
		8	
No Drug	8		
			12

Which of the following tests would be most appropriate for determining if administration of the drug increased the incidence of stomach ulcers?

- A. Analysis of variance (ANOVA)
- B. Chi-squared test
- C. Linear regression
- D. Paired t-test
- E. Pearson correlation coefficient

Explanation:

The correct answer is B. The most commonly used method for calculating p values from a two-by-two contingency table is the chi-squared test. This test is used for frequency data (such as those above) rather than for comparison of means. Since the investigator scored the stomachs as either containing ulcers or not containing ulcers, without attempting to quantify the number of ulcers, the data represent frequencies rather than means. Chi-squared is calculated as the sum for all cells of $(\text{Observed} - \text{Expected})^2 / \text{Expected}$. The p value for this value of chi-squared is obtained from a table using degrees of freedom equal to $(\text{number of rows} - 1) \times (\text{number of columns} - 1)$.

ANOVA (choice A) is used to determine if the difference between two or more groups is significantly different.

This method is not applicable to raw frequency data.

Linear regression (choice C) is the process of fitting a straight line to a set of correlational data points by minimizing the sum of squares of the vertical distances from the points to the line. This method would not be suitable for the above data.

A paired t-test (choice D) is used to compare the means of two groups. It is not used to compare raw frequency data.

Correlation is used when two variables are simultaneously measured in a sample. The Pearson correlation coefficient (choice E) is used when both the independent and dependent variable are continuous. In this case, the independent variable is not continuous (since the drug is either given or not given), and the dependent variable is also not continuous (the ulcers are either present or absent).

The data presented below compares the results of a diagnostic test in the presence and absence of a disease.

	Disease present	Disease absent
Positive test	40	5
Negative test	10	95
Total	50	100

The specificity of the test is

- A. 0.05
- B. 0.40
- C. 0.80
- D. 0.90
- E. 0.95

Explanation:

The correct answer is E. Specificity is the proportion of persons without the disease who are correctly identified by the test as being disease-free. It is given by $TN/(FP + TN)$, where TN stands for true negatives (people who

do not have the disease and test negative on the test) and FP stands for false positives (people who do not have the disease but test positive on the test). In this case, $95/(5 + 95) = 95/100 = 0.95$.

The American Diabetes Association (ADA) recently lowered the cutoff value for fasting glucose used in diagnosing diabetes mellitus from 140 mg/dL to 126 mg/dL. This reference interval change would be expected to produce which of the following alterations?

- A. Decrease the test's sensitivity
- B. Increase the test's false negative rate
- C. Increase the test's negative predictive value
- D. Increase the test's positive predictive value
- E. Increase the test's specificity

Explanation:

The correct answer is C. The negative predictive value of a test (PV-) refers to the percent chance that a normal test result is a true negative (TN; a healthy person with a normal test result) rather than a false negative (FN; a diseased person with a normal test result). It is expressed by the following formula:

$$PV- = TN/(TN + FN)$$

Similarly, the positive predictive value of a test (PV+) reflects the probability that an abnormal test result represents a true positive (TP) rather than a FP (a TP is an abnormal test result in an individual with disease, while a FP is an abnormal test result in a healthy person). PV+ is calculated using the following formula:

$$PV+ = TP/(TP + FP)$$

There is a relationship between the test's sensitivity (the ability of a test to detect disease in those who truly have the disease), specificity (ability of the test to correctly identify those without disease) and the PV- and PV+, respectively. Tests with 100% sensitivity (no FNs) and tests with 100% specificity (no FPs) automatically have a PV- and PV+ of 100%, respectively.

Changing the reference interval of a test alters its sensitivity, specificity, PV+, and PV-. In this question, lowering the upper limit of the reference interval of a fasting glucose from 140 mg/dL to 126 mg/dL increases the test's sensitivity, since a lower glucose cut-off approaches the normal value for glucose in the normal population (70-110 mg/dL). Furthermore, increasing the test's sensitivity automatically increases the test's PV-, since there are fewer FNs.

The test's sensitivity is increased (choice A) rather than decreased by the given change in the reference interval.

Since the test's sensitivity is increased, the FN rate at the new reference interval is decreased (choice B).

The test's positive predictive value (choice D) decreases, since the test's specificity, which ultimately determines its FP rate, decreases as the test's sensitivity increases. Stated another way, a greater number of normal individuals will have FP fasting glucose levels when the test's upper limit is decreased to 126 mg/dL.

A test's sensitivity is inversely related to its specificity. For example, changing the reference interval of a test to increase its sensitivity automatically lowers its specificity (choice E), since the number of FPs will increase.

Similarly, when altering a reference interval to increase a test's specificity, its sensitivity is reduced because of an increase in the number of FNs.

In summary, lowering a test's upper limit of normal increases the test's sensitivity, which decreases the FN rate, and increases the PV-. Increasing the test's sensitivity also decreases the test's specificity. Decreasing a test's specificity increases the FP rate and decreases the PV+.

